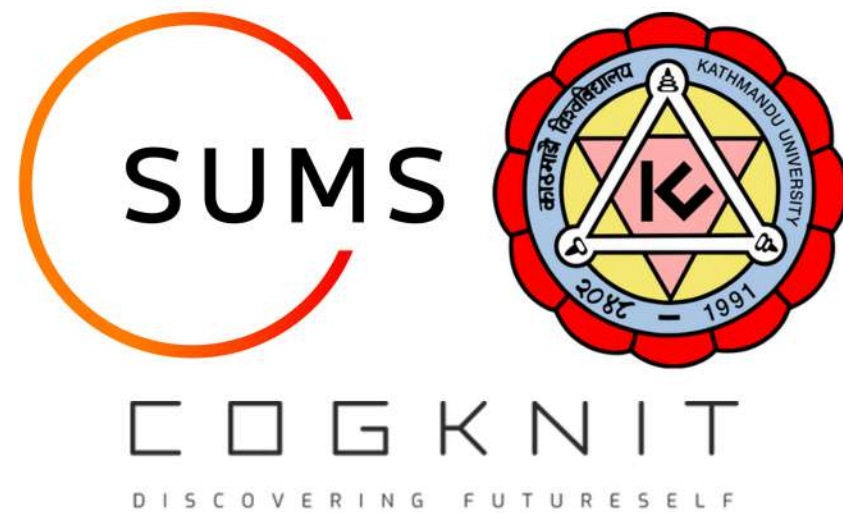




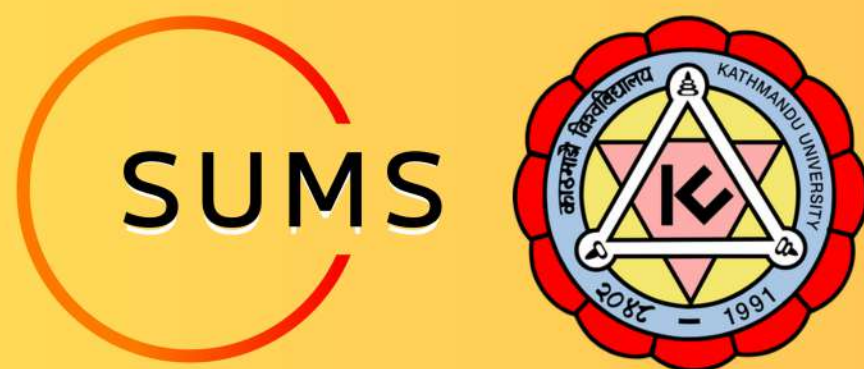
# Workshop on Smart Metering & Net Metering

IN COLLABORATION WITH  
**KATHMANDU UNIVERSITY TECHNICAL TRAINING CENTRE**  
3 DAY WORKSHOP

SEPTEMBER 2024







**"UNLEASHING THE POTENTIAL VIA  
SKILLING. RE-SKILLING & UP-SKILLING "**

# Contents

- Background
- Workshop Scope
- Methodology
- Participants
- Skill Profiling & Data Collection
- Analysing Data
- Results and Conclusion





# Background

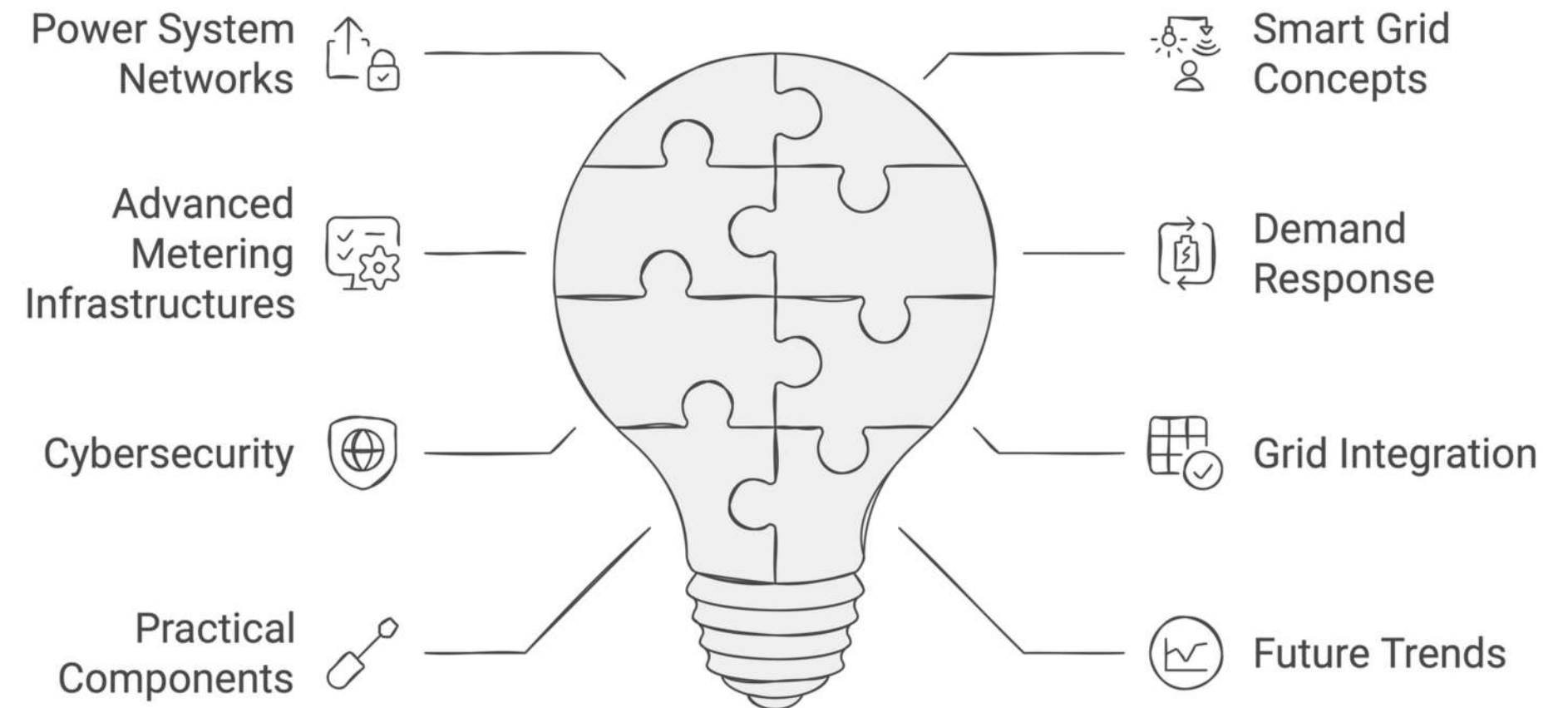


## ABOUT WORKSHOP

The 3-day workshop on Smart Metering and Net Metering, organized by Kathmandu University's Technical Training Center, provided an in-depth exploration of critical topics in the energy sector. Participants engaged with essential subjects such as power system networks, smart grid concepts, and advanced metering infrastructures (AMI and AMR). The workshop emphasized the importance of demand response, cybersecurity, and grid integration, equipping attendees with a comprehensive understanding of smart metering technologies and communication protocols.

In addition to theoretical knowledge, the workshop included practical components, such as case discussions with industrial partners and hands-on training with multifunction meters. This practical exposure allowed participants to gain valuable insights into energy auditing and load management. The event concluded with a wrap-up session that highlighted key takeaways and discussed future trends in smart metering and net metering, ensuring that attendees left with a well-rounded perspective on the evolving landscape of energy management.

## Comprehensive Overview of Smart Metering Workshop





# Workshop Scope

## SDGs

7 AFFORDABLE AND  
CLEAN ENERGY



9 INDUSTRY, INNOVATION  
AND INFRASTRUCTURE



11 SUSTAINABLE CITIES  
AND COMMUNITIES



12 RESPONSIBLE  
CONSUMPTION  
AND PRODUCTION



## Day 01

The workshop began with an introduction to power systems, covering topics like AC power, energy, voltage, frequency, and different types of meters (ToD, analog, and digital). Participants were then introduced to smart metering, discussing smart grids, AMI and AMR architectures, demand response, and distributed energy resources like solar and wind. The day also touched on cybersecurity issues in smart metering.

## Day 02

The focus shifted to smart metering architecture, technologies, communication, and data security challenges. Participants also learned about net metering, including its definition, business models, policies, and regulations. Case discussions provided real-world examples of both smart and net metering, with insights from industrial partners.

## Day 03

The final day addressed data privacy and security, exploring potential threats, protection measures, and mitigation strategies. A hands-on session followed, where participants used meters for tasks like load shedding and energy auditing. The workshop concluded with a wrap-up session summarizing key insights and next steps.







# METHODOLOGY

## Skills Profiling Based on Framework

Using the Cogknit platform, the system continuously tracked participants' performance in various internal framework skills. The platform created detailed profiles for each participant based on their self reflection across 19 key skills.

## Lecture Based Learning

Each session began with expert-led presentations on fundamental concepts such as power systems, smart metering, net metering, data privacy, and cybersecurity. Instructors used structured content to introduce technical topics and explain industry frameworks.

## Discussions and Q&A

Throughout the workshop, interactive discussions and question-and-answer sessions gauged participants' engagement, critical thinking, and problem-solving abilities, adding to their skill profiles.

## Case Studies and Real-World Applications

Case discussions, often led by industrial partners, showcased real-world applications of smart and net metering technologies. This exposed participants to practical challenges, helping to profile their ability to apply theoretical knowledge to real-life scenarios.

## Reflection and Feedback

Through the Cogknit platform, participants engaged in self-reflection and received detailed skill profiling, which offered insights into their strengths and areas for improvement. This feedback mechanism not only enhanced their learning experience but also provided a roadmap for future professional development.

The workshop concluded with a reflection session where participants discussed their learning experiences, highlighting areas of strength and improvement. The system's profiling supported personalized feedback for further skill enhancement.







# PARTICIPANTS



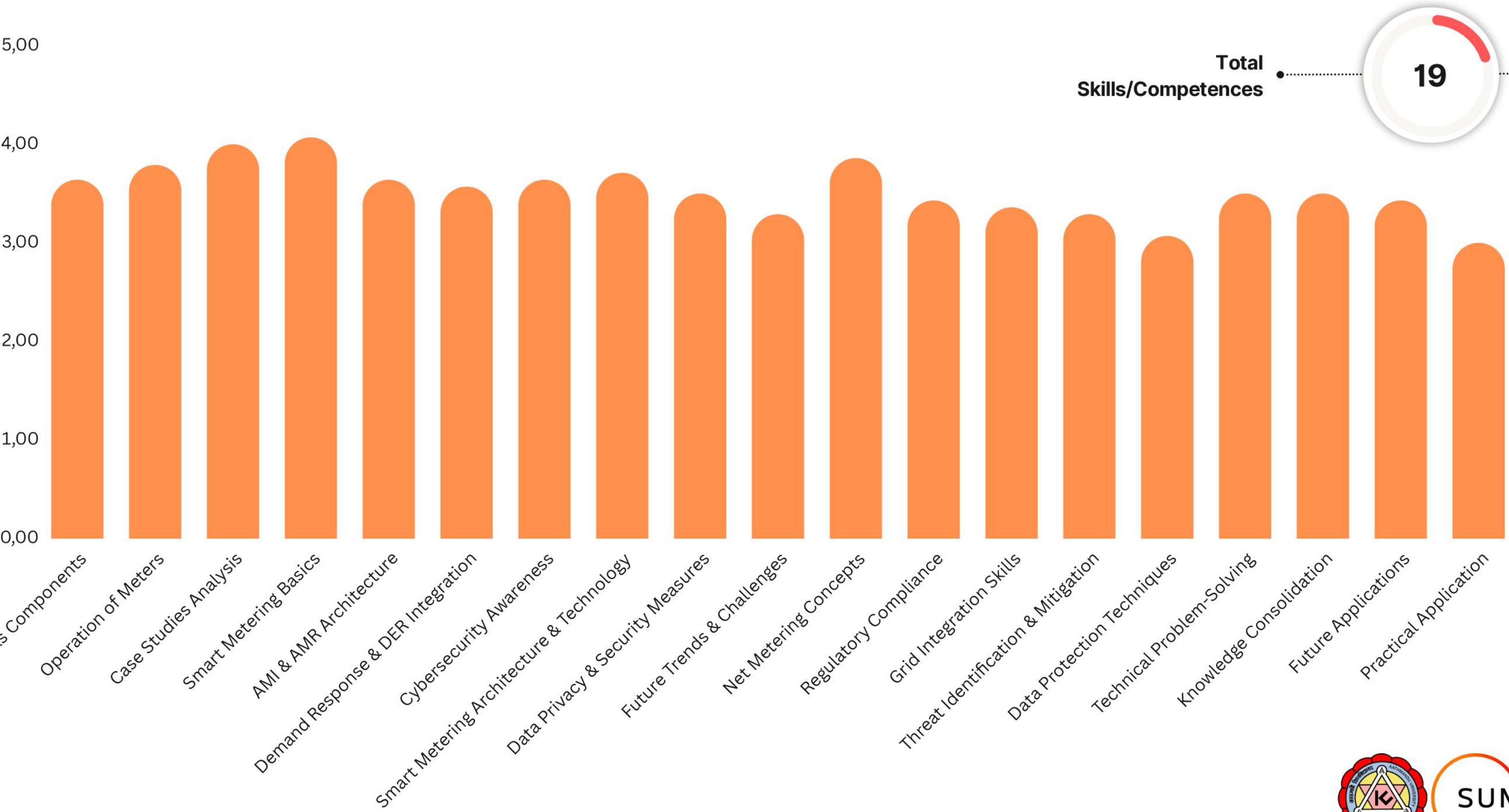
**50**  
ENROLMENTS

**90%**  
ATTENDANCE

**18%**  
**08%**  
ABSENCE



# SKILL PROFILING & DATA COLLECTION





# ANALYSING DATA

The data is analysed based on **self assessment** by the learners giving the perspectives in strengths and need to work and train in specific skills.



**3 HIGHEST**  
Overall Skills GPA



**3 LOWEST**  
Overall Skills GPA

Smart Metering Concept



Net metering concept



Case Study Analysis



Practical Applications



Data Protection  
Technique

Future Trends & Challenges  
Threat identification & Mitigation





# Results and Conclusion



- Participants showed strong analytical abilities in understanding and discussing smart metering concepts and infrastructure.
- Cybersecurity and technical problem-solving were areas of concern, requiring additional focus.

For the top 3 highest skills, such as **Case Study Analysis, Smart Metering Basics,** and **Net Metering**, Cogknit's insights enable the workshop provider to challenge participants further with more advanced content.

For the bottom 3 skills, including **Data Protection Technique , Future Trends & Challenges, Threat identification & Mitigation** and **Practical Application**, the data guides the creation of more focused, hands-on sessions to bridge knowledge gaps. By supporting a data-driven approach, Cogknit empowers the workshop provider to tailor the curriculum and improve overall learning outcomes.





# Get In Touch

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